

3.2.1 Mechanics

- **Scalars and vectors**

The addition of vectors by calculation or scale drawing. Calculations will be limited to two perpendicular vectors.

The resolution of vectors into two components at right angles to each other; examples should include the components of forces along and perpendicular to an inclined plane.

Conditions for equilibrium for two or three coplanar forces acting at a point; problems may be solved either by using resolved forces or by using a closed triangle.

- **Moments**

Moment of a force about a point defined as force $\times$ perpendicular distance from the point to the line of action of the force; torque.

Couple of a pair of equal and opposite forces defined as force $\times$ perpendicular distance between the lines of action of the forces.

The principle of moments and its applications in simple balanced situations.

Centre of mass; calculations of the position of the centre of mass of a regular lamina are not expected.

- **Motion along a straight line**

Displacement, speed, velocity and acceleration.

$$v = \frac{\Delta s}{\Delta t}, \quad a = \frac{\Delta v}{\Delta t}$$

Representation by graphical methods of uniform and non-uniform acceleration; interpretation of velocity-time and displacement-time graphs for uniform and non-uniform acceleration; significance of areas and gradients.

Equations for uniform acceleration;

$$v = u + at, \quad s = \left(\frac{u + v}{2} \right) t$$

$$s = ut + \frac{at^2}{2}, \quad v^2 = u^2 + 2as$$

Acceleration due to gravity, g ; detailed experimental methods of measuring g are not required.

Terminal speed.

- **Projectile motion**

Independence of vertical and horizontal motion; problems will be solvable from first principles. The memorising of projectile equations is not required.

- **Newton's laws of motion**

Knowledge and application of the three laws of motion in appropriate situations.

For constant mass, $F = ma$.

- **Work, energy and power**

$$W = Fs \cos \theta$$

$$P = \frac{\Delta W}{\Delta t}$$

$$P = Fv$$

$$\text{efficiency} = \frac{\text{useful output power}}{\text{input power}}$$

- **Conservation of energy**

Principle of conservation of energy, applied to examples involving gravitational potential energy, kinetic energy and work done against resistive forces.

$$\Delta E_p = mg\Delta h$$

$$E_k = \frac{1}{2}mv^2$$